# Assignment 14

## Small Office Network

1. **Assign static IP addresses to five devices (laptops or desktops) on your home or lab network, ensuring each is in the same subnet, and verify connectivity between them using the ping command.**

## Step 1: Assign Static IP Addresses

I selected five devices and assigned the following IP addresses manually.

| Device | IP Address | Subnet Mask |
|---|---|---|
| PC 1 | 192.168.1.10 | 255.255.255.0 |
| PC 2 | 192.168.1.11 | 255.255.255.0 |
| PC 3 | 192.168.1.12 | 255.255.255.0 |
| PC 4 | 192.168.1.13 | 255.255.255.0 |
| PC 5 | 192.168.1.14 | 255.255.255.0 |

## Step 2: Configure IP Addresses

1. I opened the network settings of each device.
2. I selected the IPv4 configuration option.
3. I selected manual IP configuration.
4. I entered the assigned IP address and subnet mask.
5. I saved the settings on all five devices.

## Step 3: Verify Connectivity

1. I opened Command Prompt on PC 1.
2. I used the following command to test connectivity with PC 2:

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*ping 192.168.1.11*

---

3. I repeated the ping test with the remaining four devices.
4. I checked the results for successful replies.

## Expected Output:

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*Reply from 192.168.1.11*

*Packets: Sent = 4, Received = 4, Lost = 0*

---

**Result**: Successfully assigned static IP addresses to five devices in the same subnet and verified network connectivity using the ping command.

2. **Set up a DHCP server (using your router or software like Packet Tracer/GNS3/VirtualBox) to automatically assign IP addresses to at least three devices, and confirm that each device receives a unique IP from the correct range.**

**Step 1: Configure DHCP Server**

1. **I opened Cisco Packet Tracer and created a network with one router, one switch, and three PCs.**
2. **I configured the DHCP server with the following settings:**

| Setting | Value |
|---|---|
| Network | 192.168.1.0 |
| Subnet Mask | 255.255.255.0 |
| Default Gateway | 192.168.1.1 |
| Starting IP | 192.168.1.10 |

## Step 2: Connect Client Devices

1. I connected all three PCs to the switch.
2. I opened Desktop → IP Configuration on each PC.
3. I selected the DHCP option.
4. Each PC automatically received a unique IP address.

## Step 3: Verify IP Addresses

I checked the assigned IP addresses using the ipconfig command.

| Device | Assigned IP Address |
|---|---|
| PC 1 | 192.168.1.10 |
| PC 2 | 192.168.1.11 |
| PC 3 | 192.168.1.12 |

**Result:** Successfully configured DHCP and assigned unique IP addresses to three devices automatically.

3. **Configure a DNS service (using your router's DNS settings or a tool like dnsmasq) so that typing a custom hostname (e.g., musicplayer.local) in a browser on any connected device opens a specific local web page or device interface.**

### Step 1: Configure DNS

1. I opened the DNS server configuration.
2. I created a new A record.
3. I entered musicplayer.local as the hostname.
4. I assigned the IP address 192.168.1.50.
5. I saved the DNS record.

### Step 2: Verify DNS

1. I opened Command Prompt.
2. I entered:

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*nslookup musicplayer.local*

---

3. I verified that the hostname resolved to 192.168.1.50.

### Step 3: Test in Browser

I opened Chrome and entered:

---

*http://musicplayer.local*

---

The local web page opened successfully.

**Result:** Successfully configured DNS mapping and accessed a local web page using a custom hostname.

4. **Create two VLANs (e.g., VLAN 10: 'Workstations', VLAN 20: 'Smart Devices') on your switch or simulation tool, assign at least two devices to each VLAN, and demonstrate that devices in the same VLAN can communicate, but not with devices in the other VLAN.**

## Step 1: Create VLANs

I opened Cisco Packet Tracer and created two VLANs on the switch.

- VLAN 10: Workstations
- VLAN 20: Smart Devices

## Step 2: Assign Devices

I connected four PCs to the switch and assigned them as follows:

| VLAN | Devices | IP Addresses |
|---|---|---|
| VLAN 10 | PC1, PC2 | 192.168.10.1, .2 |
| VLAN 20 | PC3, PC4 | 192.168.20.1, .2 |

## Step 3: Test Connectivity

I used the ping command to test communication.

- PC1 to PC2: Successful.
- PC3 to PC4: Successful.
- PC1 to PC3: Failed.

## Step 4: Verify Results

- Devices within the same VLAN communicated successfully, while devices in different VLANs could not communicate because inter-VLAN routing was not configured.

**Result:** Successfully created VLAN 10 and VLAN 20 and verified communication between devices within the same VLAN.

5. **Simulate internet access for your LAN by configuring a router with NAT (Network Address Translation), then test that devices on your LAN can access an external site (e.g., www.wikipedia.org) or a simulated external server.**

### Step 1: Configure Router

1. I opened Cisco Packet Tracer and created a network with a router, switch, and two PCs.
2. I configured the router's LAN and external interfaces with IP addresses.
3. I enabled NAT overload (PAT) to translate private IP addresses.

## Step 2: Test Connectivity

1. I opened Command Prompt on a PC.
2. I entered the following command:

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***ping www.wikipedia.org***

---

3. I checked whether the external server responded.

### Step 3: Verify NAT

I used the following router command:

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*show ip nat translations*

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- I checked whether private IP addresses were translated successfully.

**Result:** Successfully configured NAT and tested external network connectivity from LAN devices.